BOWIE STATE UNIVERSITY SPRING 2026 BUIS 305 / INSS 405

GROUP PROJECT WORK

Due date: May 3rd, 2026, 11:59pm

There are six defined project topics, and students are required to **choose one topic** for their project. Students may also propose their own topic, provided it is similar in scope to the six suggested topics. Project groups can have a maximum of three members, but students may also work individually if they prefer.
The project should cover all of the following areas:

- Object-Oriented Programming (OOP)
- Web development using Streamlit
- Data visualization with Pandas, Matplotlib, and Seaborn
- Data handling and cleaning
- Basic business analytics
- Python programming code

Each project topic includes recommended datasets, but students may select alternative datasets if desired. Clear submission requirements are provided for all projects.

1. Choosing a Topic

Review the suggested project topics:

1. **Retail Store Management** – Sales and customer insights dashboard
2. **Customer Segmentation**
3. **Banking & Finance** – Transaction and fraud analysis dashboard
4. **Employee & HR** Analytics
5. **Education** – Student performance and attendance dashboard
6. **Public Health** – Patient and hospital data dashboard

2. Completing the Project

- **Step 1: Data Preparation**
  - Load the dataset in Python using Pandas.
  - Clean the data (handle missing values, format dates, remove duplicates).
- **Step 2: Implement OOP**
  - Create at least **one class** with methods for:
    - Data cleaning and preprocessing
    - Data analysis and aggregation
    - Visualization of charts
- **Step 3: Build the Streamlit Dashboard using Python**
  - Include interactive charts and filters relevant to your chosen topic.

  Examples:
  - Retail: Sales by product, customer types, or region
  - Banking: Transactions over time, anomaly detection
  - Education: Grades distribution, attendance analysis

        - Public Health: Patient outcomes
- **Step 4: Analysis and Insights**
    - Summarize your findings in clear, actionable statements.
    - Include charts in your report to support your analysis.

3. Submission Requirements

Each student must submit:

- **Python Code File**: Well-commented app.py implementing OOP and Streamlit dashboard **(20 points)**
- **Project Report** (2-4 pages):
    - Objective of the project
    - Data cleaning and preprocessing steps **(5 points)**
    - Methods and analysis
    - Key findings and insights **(20 points)**
- **Outputs**:
    - Screenshots of dashboard charts and summaries,
    - Visualizations that support findings **(20 points)**
- **PowerPoint Presentation** (8–12 slides):
    - Introduction & dataset description
    - Methods
    - Visualizations and insights
    - Conclusion **(20 points)**
- **Any additional supporting documents**
    Dataset (if modified)

4. Submission Procedure

1. Upload your Python code and all required files to the designated assignment link.
2. Ensure all files are correctly named, e.g., YourName_ProjectTopic.py. or AllGroupMembersLastNames_ProjectTopic.py eg, BossWood_ProjectTopic.py
3. Double-check that your dashboard works on **Streamlit Cloud** (share.streamlit.io) and include the **URL** in your submission. (**15 points**)

<u>PROJECT TOPICS</u>

Project 1

Retail Store Management: Sales & Customer Insights Dashboard

Objective: Build an interactive web dashboard to analyze sales performance and customer behavior.

Project Questions:

1. Which products or categories generate the most revenue?
2. How do sales vary over time (daily, monthly)?
3. Who are the top customers by total spending?
4. What is the distribution of new vs returning customers?
5. Which region or store location performs best?

Instructions:

1. Import a sales dataset (CSV or Excel) using Streamlit file uploader.
2. Clean missing values and inconsistent date formats.
3. **Create charts for**:
   - Sales by product category (bar chart)
   - Sales by region or store (bar chart)
   - sales vary over time (daily, monthly) line chart
   - Top customers by total spending
   - Distribution of new vs returning customers – Pie Chart
4. Use OOP to create a class SalesAnalyzer with methods for cleaning, summarizing, and visualizing data.
5. Deploy the final dashboard using GitHub + Streamlit Cloud.

Applied Methods

- Aggregation by product/category
- Visualization with bar, line, and pie charts
- Streamlit interactivity (filters by product, region, date)

Suggested Dataset:

- Superstore Sales Dataset - sample-superstore, sample-data
- Walmart Sales Forecasting Dataset - Walmart Recruiting - Store Sales Forecasting | Kaggle, walmart-sales-data
- Online Retail (UCI) - Online Retail, online-retail-dataset

Project 2: Customer Segmentation & Behavior Insights

Objective: Analyze customer purchasing behavior and create a dashboard for business decisions.

Instructions:

1. Clean and preprocess customer demographic and purchase datasets.
2. Use OOP to build a CustomerAnalysis class for:
   - Descriptive statistics
   - Filtering new vs returning customers
   - Total spending per customer
3. Create visualizations showing:
   - Purchase frequency
   - Average spending by customer type
   - Customer distribution by age or region
4. Create a Streamlit dashboard with at least 3 interactive filters (date, region, segment).

Suggested Dataset:

- Online Retail Dataset - Online Retail. online-retail-dataset
- Customer Personality Analysis Dataset - Customer Segmentation, Customer Personality Analysis
- Mall Customer Segmentation Dataset - Mall Customer Segmentation Data

Project 3: Employee Performance & HR Analytics Dashboard

Objective: Build an HR analytics tool with visual insights.

Instructions:

1. Load and clean HR employee data.
2. Create an OOP class HRDashboard for:
   - Removing duplicates
   - Summaries (attrition rate, avg. salary, performance score)
   - Visualizations
3. Create charts:
   - Attrition by department
   - Salary distribution
   - Performance rating distribution
4. Build a Streamlit web app with interactive department filters.

Suggested Dataset:

- IBM HR Analytics Employee Attrition Dataset - IBM HR Analytics Employee Attrition & Performance, employee-attrition-sample-dataset
- HR Employee Performance Dataset - employee-performance

Project 4

Banking & Finance: Financial Transaction & Fraud Analysis

Objective: Analyze financial transactions to detect anomalies or unusual behavior and summarize customer activity.

Project Questions:

1. What is the total and average transaction amount per customer?
2. Detect unusually high-value transactions (potential fraud)?
3. What are spending patterns by category or account type?
4. How do transactions trend over time?
5. Visualize new vs existing customers' transaction behavior.

Instructions:

1. Clean financial transactions dataset.
2. Build a TransactionAnalyzer class with OOP methods:
   - Cleaning transaction data
   - Summary of total spending
   - Detection of outliers (anomalies or unusual transactions) using simple statistical rules (e.g., 3σ rule - values outside mean $\pm 3 \times$ standard deviation)
   - Aggregating and visualizing customer spending
3. Visualizations:
   - Transactions overtime
   - High-value vs normal transactions
   - Category-wise spending
4. Streamlit dashboard should allow users to upload files and analyze transactions.

Suggested Dataset:

- Credit Card Transaction Dataset - Credit Card Fraud Detection , paysim
- Bank Marketing Dataset - bank marketing , Bank Marketing
- Synthetic financial transaction dataset - paysim1, synthetic-financial-transactions

Methods Applied:

- Data cleaning and handling missing values
- Grouping and aggregations with Pandas
- Visualizing distributions, time series, and outliers
- Streamlit dashboard with upload and filter options

Project 5

Education: Student Performance & Attendance Dashboard

Objective: Build a dashboard to monitor student performance and attendance for decision-making.

Project Questions:

1. What is the grade distribution by class, subject, or semester?
2. How does attendance correlate with student performance?
3. Which students are at risk of failing or dropping out?
4. Compare performance between different schools or programs.
5. Visualize trends in student performance over semesters.

Instructions:

Create a class StudentAnalyzer with methods for:

- Cleaning grades and attendance data
- Computing statistics like average grades and attendance rates
- Visualizing performance trends and at-risk students

Suggested Datasets:

- Student Performance Dataset (UCI) - Student+Performance, Students' Academic Performance Dataset
- High School/College Academic Records Dataset - school-student-daily-attendance
- Kaggle Datasets: Student Scores and Attendance

Methods Applied:

- Data cleaning and missing value handling
- Grouping by student, subject, or class
- Visualizations: bar charts, scatter plots, heatmaps
- Streamlit interactivity with filters for class, subject, or date

Charts to Create:

- Bar chart: Grade distribution by subject
- Scatter plot: Attendance vs Performance
- Line chart: Average grades over semesters

Project 6

Public Health: Patient & Hospital Data Dashboard

Objective: Analyze hospital or public health data for operational and policy insights.

Project Questions:

1. What is the distribution of patient outcomes (discharge, DAMA, death) by age or gender?
2. How many admissions occur per month or year?
3. Which diseases or conditions are most common in the dataset?
4. How does length of stay vary by department or patient type?
5. Visualize service satisfaction or patient feedback ratings.

Suggested Datasets:
- Hospital Patient Records Dataset – Hospital Admissions Dataset used in class
- Public Health Surveys / CDC Data - [Data | Centers for Disease Control and Prevention]
- COVID-19 or Health Metrics Datasets (Kaggle) - [COVID-19 Open Research Dataset Challenge (CORD-19)]

Methods Applied:
- Data cleaning and handling missing or categorical data
- Grouping and aggregating outcomes
- Visualizations: histograms, bar charts, line charts
- Streamlit dashboards with interactive filters by age, gender, or department

Instructions:

Create a class HealthAnalyzer with methods for:
- Cleaning patient records
- Summarizing outcomes (discharge, DAMA, death)
- Aggregating data by age, gender, or department

Charts to Create:
- Histogram: Patient outcomes by age
- Line chart: Admissions over time
- Bar chart: Average service satisfaction by department